

MOIL AI/ML MINING INTELLIGENCE PLATFORM

Jury Presentation & Rapid Technical Reference Cheat Sheet — SIH 2026

HIGH-YIELD PRESENTATION SUMMARY

1. *Rapid Pitch Scripts (30s, 1m, 3m)*
2. *Key Architectural Mappings & Model Justifications*
3. *30 Technical Jury Q&As for Presenters*
4. *'Claims We Must Not Make' Defensive Compliance Checklist*

1. Elevator Pitch Scripts

30-Second Pitch

The MOIL AI Mining Intelligence Platform is an enterprise-grade operational suite designed specifically for Manganese Ore India Limited. It unifies underground and opencast borehole core assays, daily HEMM fleet telemetry, IMD monsoon weather risks, and satellite thermal observations into a single AI-driven system. Powered by Random Forest grade prediction and Gradient Boosting production forecasting, it enables shift managers to optimize ore blending and mitigate pit inundation before costly disruptions occur.

1-Minute Pitch

Manganese ore mining presents unique challenges in grade heterogeneity, pit slope stability, and strict steel plant dispatch specifications. Our platform addresses these challenges by transforming raw operational data into actionable shift intelligence. Using scikit-learn models trained on geological borehole core data, our system predicts manganese ore grades while strictly eliminating target leakage. Our Gradient Boosting regressor forecasts shift production based on fleet downtime and rainfall intensity. Simultaneously, our 4-pillar risk engine evaluates operational, geological, environmental, and safety risks in real time. Combined with NASA FIRMS thermal telemetry and an interactive GIS spatial workspace, the platform equips MOIL leadership with predictive decision support to maximize yield and ensure operational safety.

3-Minute Pitch

Respected Members of the Jury, MOIL Limited produces over 1.3 million tons of manganese ore annually across 11 key mines like Balaghat and Dongri Buzurg. However, mining operations face daily operational friction: sub-optimal ore blending leads to dispatch penalty rejections, heavy monsoon rainfall causes open-pit flooding, and equipment breakdowns stall shift production targets. We built the MOIL AI/ML Mining Intelligence Platform to solve these exact problems through a robust 4-layer architecture: First, our Data Quality & ETL Engine continuously audits raw observations across 5 quality domains—completeness, timeliness, validity, consistency, and accuracy. Second, our Dual AI/ML Subsystem delivers precise predictions: a Random Forest classifier categorizes borehole cores into High, Medium, Low, or Waste grade classes without target leakage, while a Gradient Boosting regressor models shift tonnage output under varying fleet breakdown and weather conditions. Third, our GIS & Remote Sensing Telemetry Layer overlays GeoJSON mine boundaries with Sentinel-2 NDVI/NDWI moisture indices and NASA FIRMS thermal hotspot warnings within a 5-kilometer perimeter. Finally, our Prescriptive Decision Support & What-If Simulator allows shift managers to simulate monsoon impact, adjust cutoff grades, and approve auto-generated blending recommendations via an audited workflow. The entire prototype has been rigorously verified with 87 automated integration tests and built using a modern stack of FastAPI, React 18, Leaflet GIS, and SQLAlchemy.

2. Tech & ML Selection Cheat Sheet

Technology / Model	Role	Key Implementation Rationale
React 18 + Vite	Frontend Client	Modular UI components, 3D Globe landing, Warm-Ivory (#F4F3EF) industrial theme.
FastAPI + Pydantic	REST API Gateway	Asynchronous Python backend, strict DTO validation, 13 REST API routes.
SQLAlchemy + SQLite	Database Engine	9 relational tables, ACID transactional support, PostGIS/PostgreSQL ready.
RandomForestClassifier	Reserve ML Model	Classifies borehole assays into 4 grade tiers without target Mn% leakage.
GradientBoostingRegressor	Production Forecast ML	Predicts shift actual tonnage output accounting for downtime & rainfall.
Leaflet 1.9 + GeoJSON	GIS Spatial Engine	Renders 8 mine polygons, 37 production zones, and Kriging grade heatmaps.

3. Defensive Compliance Checklist

 CLAIMS WE MUST NOT MAKE — JURY AUDIT

-  **DO NOT SAY:** 'We are connected live to MOIL's internal SCADA database.'
-  **SAY INSTEAD:** 'The prototype uses calibrated synthetic data modeled on MOIL published parameters with ready-to-connect API adapter contracts.'

-  **DO NOT SAY:** 'Satellite imagery directly detects underground manganese ore.'
-  **SAY INSTEAD:** 'Satellite sensors monitor surface vegetation, water accumulation, and thermal hotspots. Underground ore is analyzed via core assays.'

-  **DO NOT SAY:** 'Our ML model is 100% accurate.'
-  **SAY INSTEAD:** 'Our model achieves robust validated performance on core features while strictly preventing target leakage.'